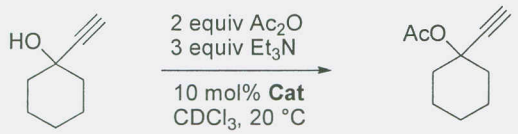
Organic Cume

10-24-08

1. 4-(Dimethylamino)pyridine (DMAP, **1**) has been widely used as a nucleophilic group-transfer catalyst for a number of applications such as acylation, esterification, and silylation, to just name a few. Recently, Steglich and his group developed a more active DMAP derivative **3** for the acylation (*Angew. Chem. Int. Ed.* **2003**, *42*, 4826-4828). The activities of DMAP (**1**) and its derivatives (**2** and **3**) were tested on the acetylation of 1-ethynylcyclohexanol (eq 1). These reactions were monitored by ^{1}H NMR spectroscopy. Their half-lives are summarized below. Please provide a detailed mechanism for the reaction shown in eq 1 and your rational for the enhanced catalytic activity of **3**.

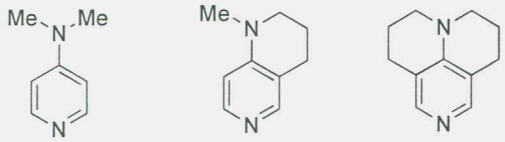


(1)

$t_{1/2}$ for the reaction shown in equation 1

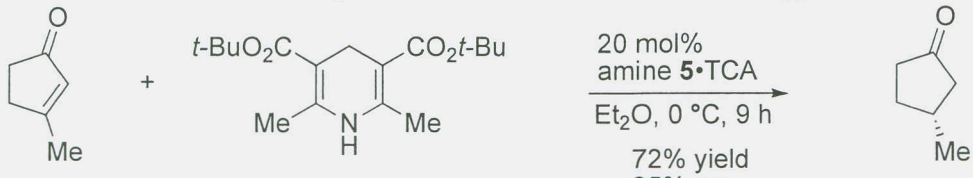
Cat	**1** (DMAP)	**2**	**3**
$t_{1/2}$ [min]	151	63	26

Cat:



1 **2** **3**

2. Organocatalysis is playing an increasingly important role in asymmetric synthesis. One of the authorities in this field, David W. C. MacMillan, reported an enantioselective organocatalytic transfer hydrogenation of cyclic enones that allows access to enantiomerically enriched β–substituted cyclic ketones using Hantzsch ester **4** (*J. Am. Chem. Soc.* **2006**, *128*, 12662-12663). Please provide a detailed mechanism for the reduction (eq 2) and 3-dimensional structures of key intermediates in this process to account for the sense of asymmetric induction. What is the byproduct derived from **4**?



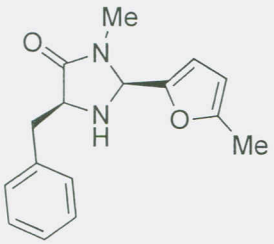
(2)

1 equiv

4, 1.1 equiv

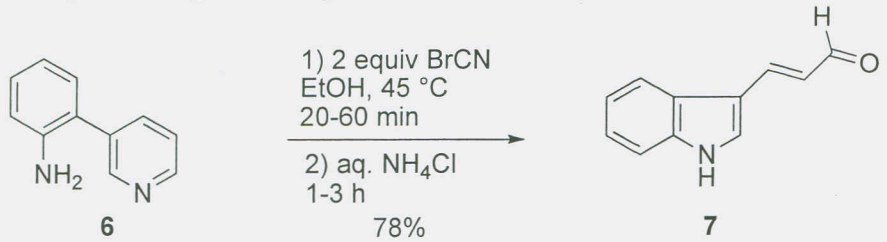
72% yield
95% ee

amine **5**:



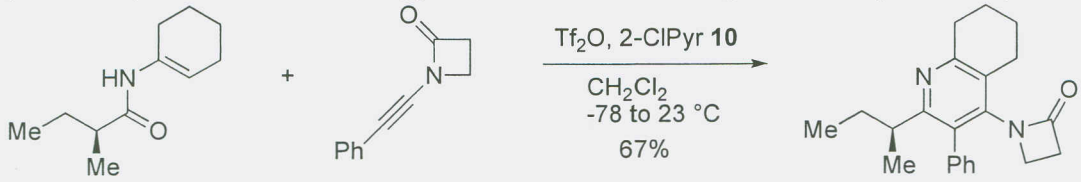
Note: TCA refers to trichloroacetic acid.

3. Vanderwal and his group recently reported a method for the synthesis of substituted indoles (e.g., **7**) from pyridinyl anilines such as **6** (*Angew. Chem. Int. Ed.* **2006**, *45*, 7803-7806). Please provide a plausible mechanism for the reaction shown in eq 3.



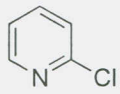
(3)

4. In 2007, Movassaghi and his group published a convergent and single–step procedure for the synthesis of highly substituted pyridines by condensation of *N*-vinyl amides with π-nucleophiles such as alkoxy and silyloxy acetylenes as well as enol ethers (*J. Am. Chem. Soc.* **2007**, *129*, 10096-10097; *J. Am. Chem. Soc.* **2006**, *128*, 14254-14255). One of the examples from this paper (**8** to **9**) is shown in eq 4. Please provide a plausible mechanism for the transformation shown in eq 4. Hint: 1) 2-ClPyr **10** was found not to add to Tf_2O; 2) react-IR monitoring during the treatment of amides with Tf_2O and 2-ClPyr **10** in the absence of π-nucleophiles revealed the consumption of 2-ClPyr **10** ($1580cm^{-1}$) with concomitant appearance of a new absorption band ($1600cm^{-1}$).



(4)

8, 94% ee

9, 94% ee

2-ClPyr **10**: